Assignment 7

M 148 - Experience of Data Science

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May 14, 2024

This assignment is due by the end of the day on Tuesday of week 8. This is a group assignment, it is your group leader's responsibility to upload it to BruinLearn by the deadline. Please make sure you upload a pdf.

Task 1 - time series of number of orders shipped

Create a time series for the number of orders shipped for your chosen level of aggregation (daily / weekly / monthly). Tell me about the model you are using for your data set, why did you chose it? What coefficients are you including and why? Plot or tabulate the coefficients and interpret them. Create a forecast for 2024 and submit a plot of the predictions.

For the predictions, please report your predicted mean (this will be used to compute your predictive accuracy using MSE) and a 95% confidence interval.

Task 2 - improve predictive performance of your model for whether an incomplete journey will have an order shipped

Do you need to adjust your strategy for creating a training data set? Are there additional predictor variables you want to compute? Should you adjust the way you compute your training data? Start working on improving your predictive power for the second test data set. You do not need to provide any code or plots, just tell me about what you are improving.

Task 3 - plot predicted probability over time

Create plots that shows a single journey's predicted probability of success over time. Make sure you include all user actions in your plot. Feel free to use Russel_f1.png and Russel_f2.png as templates. Improve on them by making sure the actions that trigger an increase are correctly placed before the increase.

Task 4 - dig deeper continued

Find a new interesting question or keep working on your previous question(s). Let me know how things are going. What are problems you have been facing, what are (preliminary) results? Good topics include:

- Can we use an LSTM to predict which journeys will end in a success and which ones will end in a failure? It seems suitable, because it can naturally take inputs of arbitrary length, but I am not sure how to handle the differing time intervals between events (maybe as an additional predictor?). A maybe even bigger problem is how to tell the model that a journey is still ongoing and how much time has elapsed since the last action.
- When do customers drop out? In terms of time, actions, milestones reached, etc.
- What is the effect of promotions? Can you estimate some sort of causal effect of promotions on user behavior?
- Create a Markov chain analysis for modeling user behavior, model transition probabilities between events. What is typical use behavior? Can you include wait-times between events?
- Can you do user segmentation beyond successful vs. unsuccessful journeys? Clustering might be a useful tool. Are you able to cluster incomplete journeys or make probabilistic predictions into which cluster (of completed journeys) they will fall?

This is really open-ended. Anything goes, it's time to branch out.

10pts total